

Trained

Place labels decide what counts

The diagram illustrates the training process. On the left, a stack of documents is shown, each containing a grid of small bar charts representing different features. An arrow labeled 'train' points to the right, where a single document is shown with a grid of colored squares (blue, orange, purple) representing the extracted features.

Large labelled training set

Corpus-specialized features

Standard	Strong within-domain discrimination
Cross-env.	Can inherit training-domain bias

The diagram illustrates the process of training a model. On the left, a stack of documents is labeled "Large labelled training set". An arrow labeled "train" points to the right, leading to a box containing several colored squares (blue, purple, brown) labeled "Corpus-specialize features".

Added criteria decide what counts

Standard	Selection ≠ place discrimination
Cross-env.	Selection criteria may not transfer

Ours

Let patches decide what counts

The diagram illustrates the 'Ours' method. It starts with a stack of 'Frozen patch features' (represented by a 4x4 grid of blue squares). An arrow points to a single 4x4 grid labeled 'Within-image' and 'Balanced, matchable patch evidence', which shows a path of four blue squares. To the right, another 4x4 grid labeled 'Across-image' shows a path of four squares, with two blue squares on the left and two purple squares on the right, connected by green lines. This represents matching evidence across different environments.

Standard	Match distinctive local evidence
Cross-env.	Strong cross-environment retrieval

The diagram illustrates the process of generating balanced, matchable patch evidence from frozen patch features. It is divided into three main stages:

- Frozen patch features:** On the left, a stack of three 4x4 grids represents frozen patch features. Each grid contains four blue squares at positions (1,1), (1,3), (3,1), and (3,3) (assuming 0-indexing from top-left).
- Within-image:** An arrow points to a single 4x4 grid. This grid shows the same four blue squares, but now each has a green line connecting it to its corresponding square in the same grid, representing self-evidence or within-image matching.
- Across-image:** An arrow points to two 4x4 grids. The left grid is identical to the 'Within-image' grid. The right grid is a 4x4 grid with four purple squares at the same positions (1,1), (1,3), (3,1), and (3,3). Green lines connect the blue squares in the left grid to the purple squares in the right grid, representing cross-image matching evidence.

Below the diagrams, the text reads: "Within-image" and "Across-image" in green, followed by "Balanced, matchable patch evidence" in green.

Standard	Match distinctive local evidence
Cross-env.	Strong cross-environment retrieval

VP-Air R@1 (cross-environment benchmark, %)

